

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Ramadhan Pratama Siregar	Indonesia	rpsrg1@gmail.com	
Ruthvik Mohan Menduguttala	India	ruthvikmohan007@gmail.com	
Ozioma Okoroafor	Nigeria	ozimaokoroafor@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

Team member 1	Ramadhan Pratama Siregar
Team member 2	Ruthvik Mohan Menduguttala
Team member 3	Ozioma Okoroafor

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Executive Summary

This project models 25-day ahead returns for SPY, TLT, SHY, GLD, and DBO. This report evaluates five ETFs across equities, fixed income, and commodities using three single-output deep learning models and a final multi-output LSTM. The forecasts are then converted into a simple allocation rule to test whether predictive performance can translate into better portfolio results.

The final strategy selected the top two predicted ETFs every 25 trading days. In the test period, the equal-weight benchmark outperformed the model-driven strategy, showing that prediction quality did not consistently translate into superior allocation performance.

Methodology and Dataset

Daily adjusted close prices were converted into returns, and each target was the 25-day ahead return. Single-output models used 60 lagged daily returns, while the multi-output LSTM used a 60-day sequence across all ETFs. This is a supervised regression setup (Zhang et al., 2023).

Table M.1. Modelling setup used across all ETFs.

Item	Description
Assets	SPY, TLT, SHY, GLD, DBO
Common price sample	4,025 daily observations
Modelling data	3,999 observations after returns and target alignment
Input window	60 lagged daily returns
Forecast horizon	25 trading days
Training period	Before 2016
Validation period	2016-2017
Test period	2018-2022

The chronological split reduces look-ahead bias because later market data is not used to train earlier predictions. Returns were used instead of price levels to reduce nonstationarity before model fitting (Hamilton, 1989).

Step 1: Data Gathering and Exploratory Analysis

The exploratory analysis compares broad equity, fixed income, cash-like, precious metal, and crude oil exposures. This common view supports later team modelling because the multi-output LSTM depends on cross-asset information rather than one asset alone.

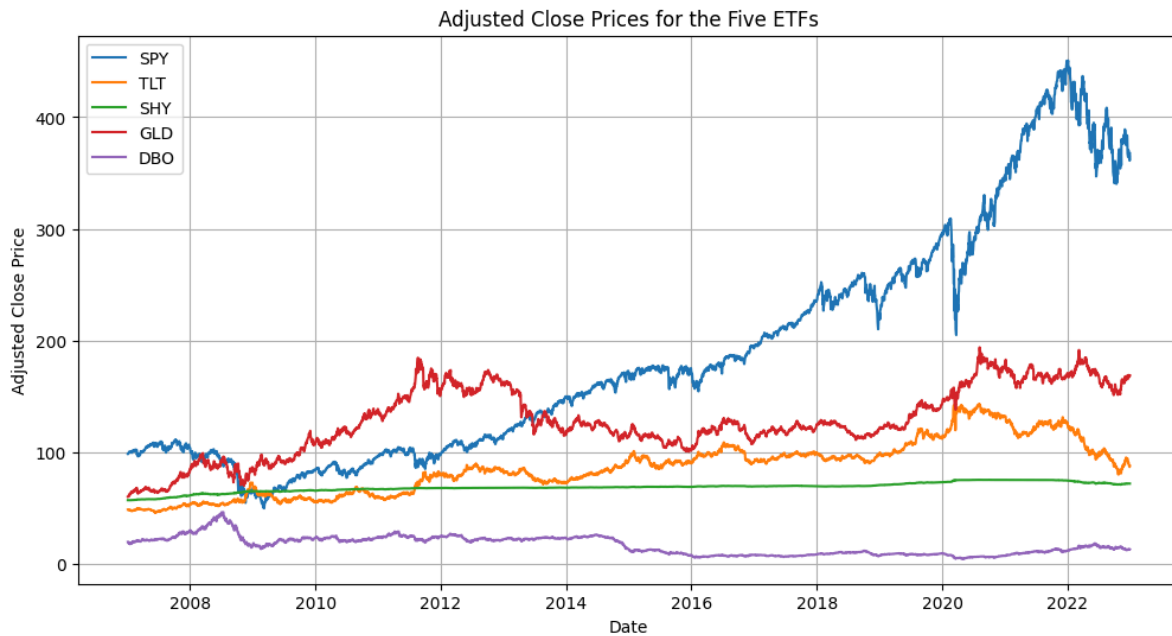


Figure E.1. Adjusted close prices for SPY, TLT, SHY, GLD, and DBO.

SPY shows the strongest long-term price growth, while SHY is comparatively flat. GLD and DBO behave differently from equity and bond ETFs, supporting their inclusion as commodity-related diversifiers.

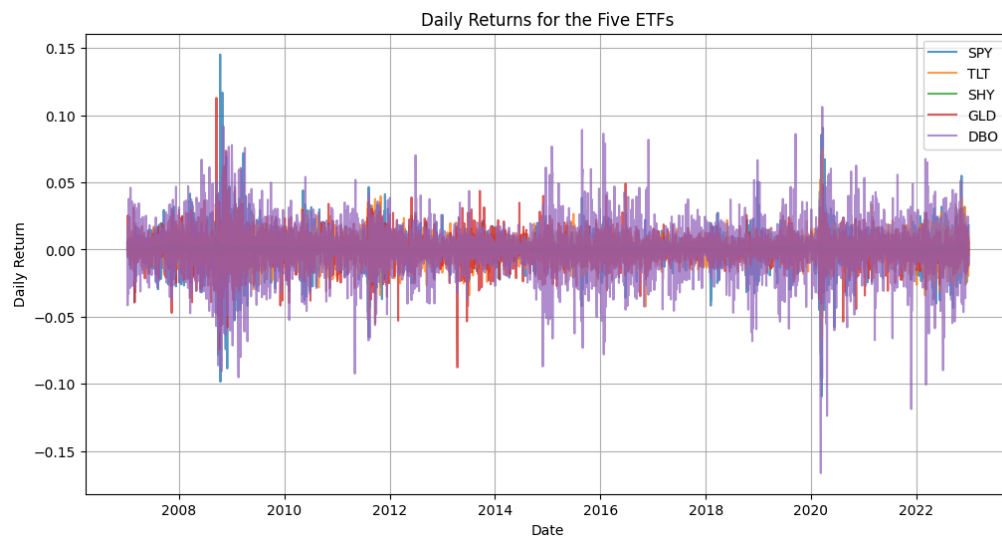


Figure E.2. Daily returns for all five ETFs.

Returns fluctuate around zero, but DBO has the largest spikes. SHY has the smallest daily movements, making it the most stable asset in the sample.

Table E.1. Summary statistics and ADF test results for daily returns.

Ticker	Mean Daily Return	Daily Volatility	Skewness	Kurtosis	ADF Statistic	ADF p-value
SPY	0.000411	0.012905	-0.064613	13.638969	-15.761273	1.19e-28
TLT	0.000192	0.009582	0.099062	3.713499	-12.129646	1.75e-22
SHY	5.80e-05	0.000882	0.011940	7.686719	-9.631119	1.61e-16
GLD	0.000319	0.011152	-0.108977	6.781592	-63.840401	0.000000
DBO	0.000102	0.020110	-0.381939	3.994558	-66.613672	0.000000

All return series have very small ADF p-values, supporting stationarity after conversion from prices to returns.

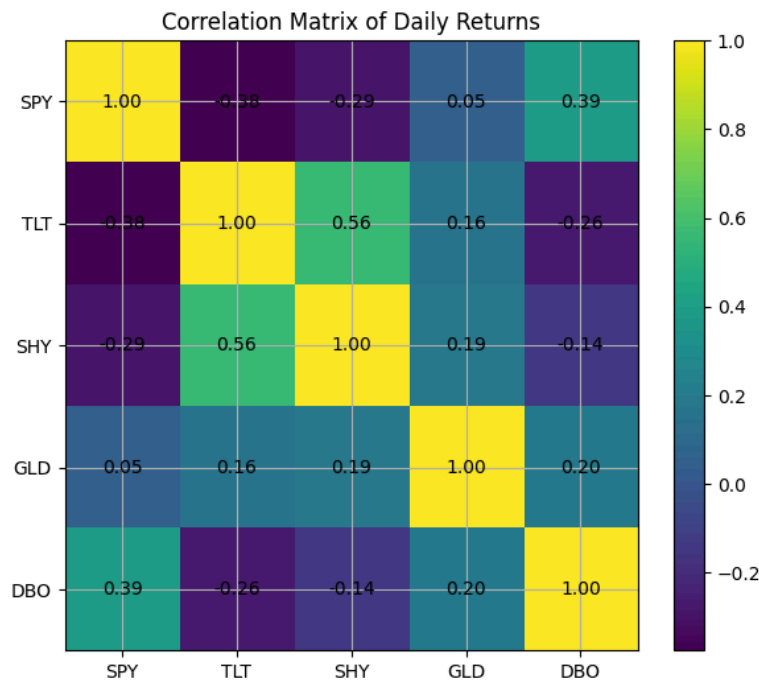


Figure E.3. Correlation matrix of daily returns.

SPY is negatively correlated with TLT and positively correlated with DBO. TLT and SHY move together, while GLD has weaker links with the other ETFs.

Table E.2. Correlation matrix for daily ETF returns.

Ticker	SPY	TLT	SHY	GLD	DBO
SPY	1.000000	-0.375294	-0.292188	0.049565	0.391156
TLT	-0.375294	1.000000	0.557702	0.163450	-0.264109
SHY	-0.292188	0.557702	1.000000	0.192602	-0.141337
GLD	0.049565	0.163450	0.192602	1.000000	0.203552
DBO	0.391156	-0.264109	-0.141337	0.203552	1.000000

Section 1: SPY - Equity ETF

SPY represents the broad U.S. equity market in the group portfolio. Its inclusion adds an equity-risk signal to the fixed-income and commodity assets. The same 60-day input window and 25-day target were used for consistency.

1.1 Data Behaviour

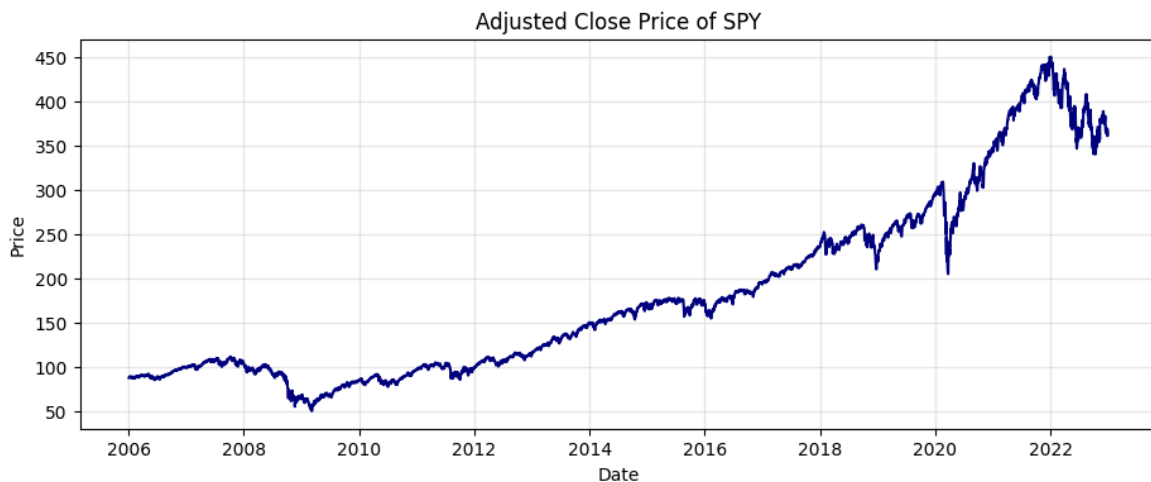


Figure 1.1. Adjusted close price of SPY.

SPY trends upward over the full sample, with visible drawdowns around major stress periods. This makes raw prices unsuitable as stationary model inputs.

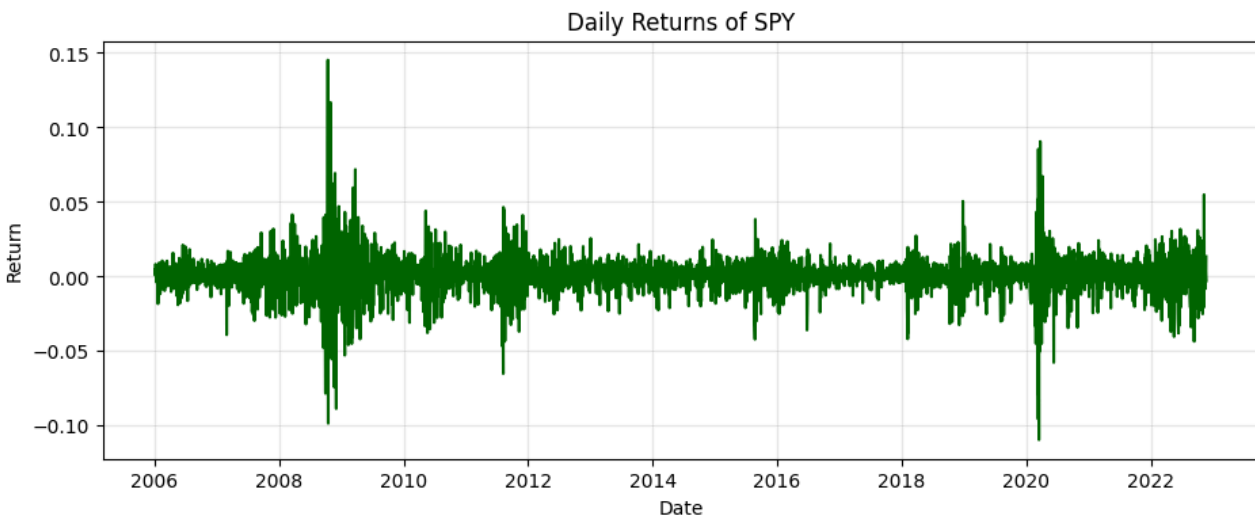


Figure 1.2. Daily returns of SPY.

SPY returns cluster around zero, but volatility rises around crisis periods. The 2020 shock is especially visible in the return series.

1.2 Single-Output Models

The MLP, CNN-GAF, and LSTM models were trained separately for SPY. The MLP provides a nonlinear benchmark, CNN-GAF uses an image transformation, and LSTM treats the 60-day return window as a sequence.

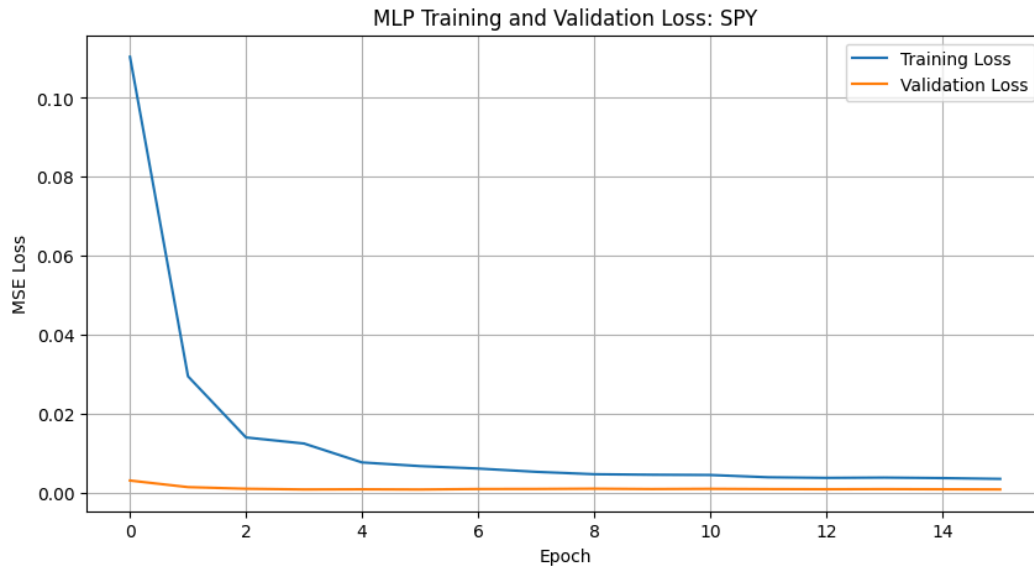


Figure 1.3. MLP training and validation loss for SPY.

The SPY MLP loss declines steadily, while validation loss remains lower and stable. This indicates stable fitting, but not necessarily strong forecasting power.

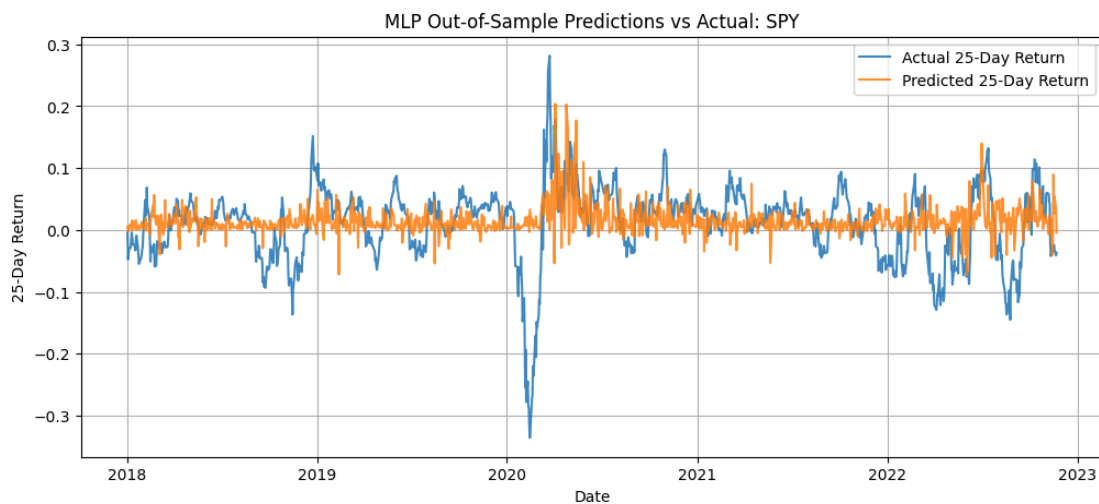


Figure 1.4. MLP out-of-sample predictions for SPY.

MLP predictions are smoother than actual SPY returns. The model captures only limited amplitude during large market moves.

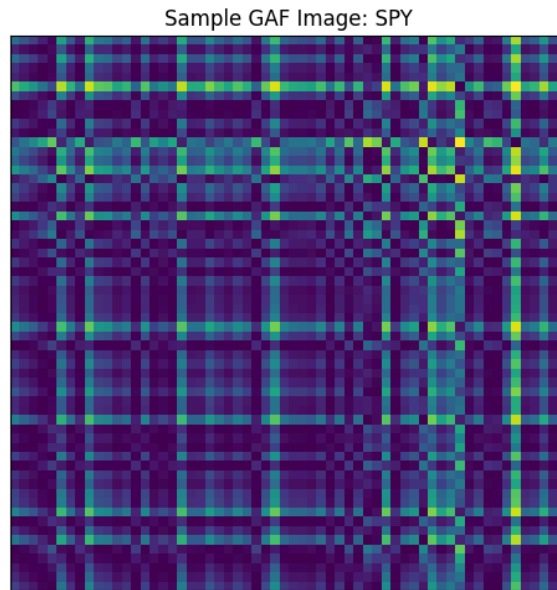


Figure 1.5. Sample GAF image for SPY.

The GAF image converts a 60-day SPY return window into a two-dimensional pattern. CNNs then search this representation for local structures.

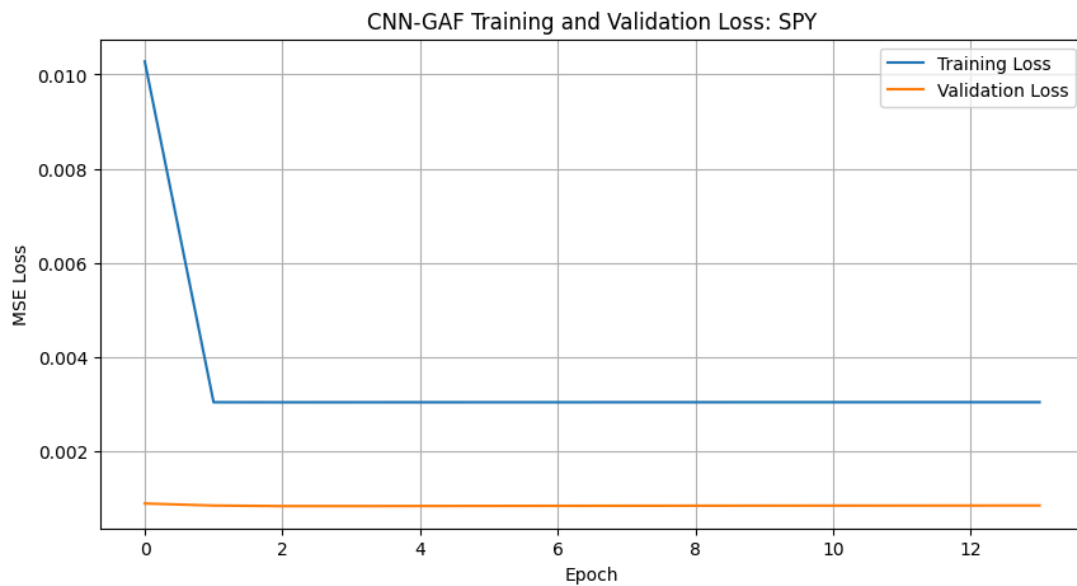


Figure 1.6. CNN-GAF training and validation loss for SPY.

The CNN-GAF loss drops quickly and stabilizes. This suggests the CNN fitted the transformed image inputs early in training.

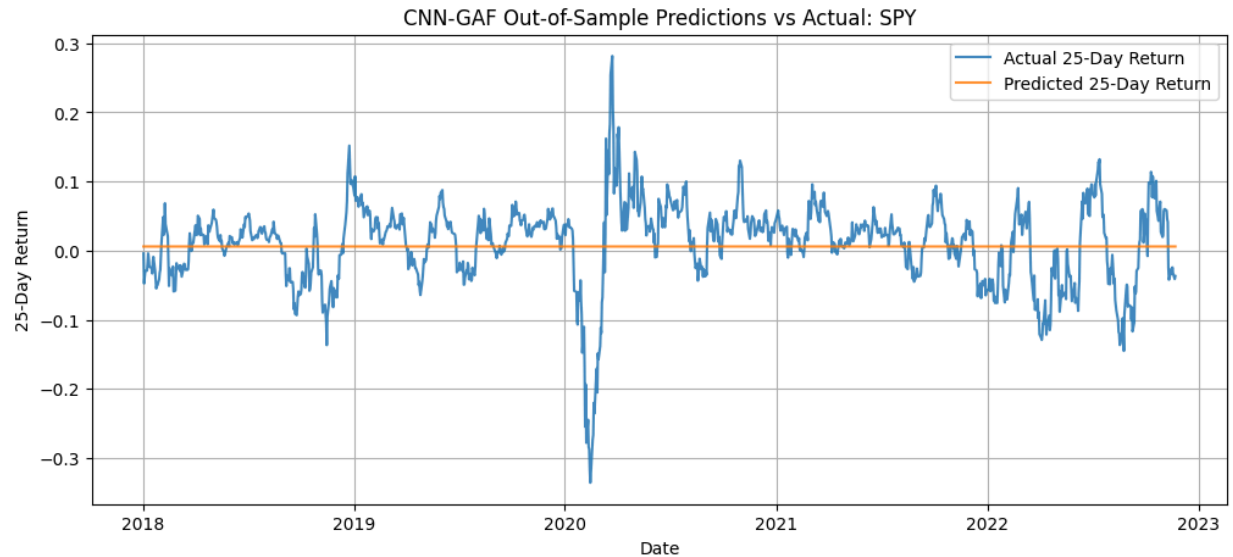


Figure 1.7. CNN-GAF out-of-sample predictions for SPY.

CNN-GAF predictions remain smoother than actual returns. The model improves directional accuracy but still misses large swings.

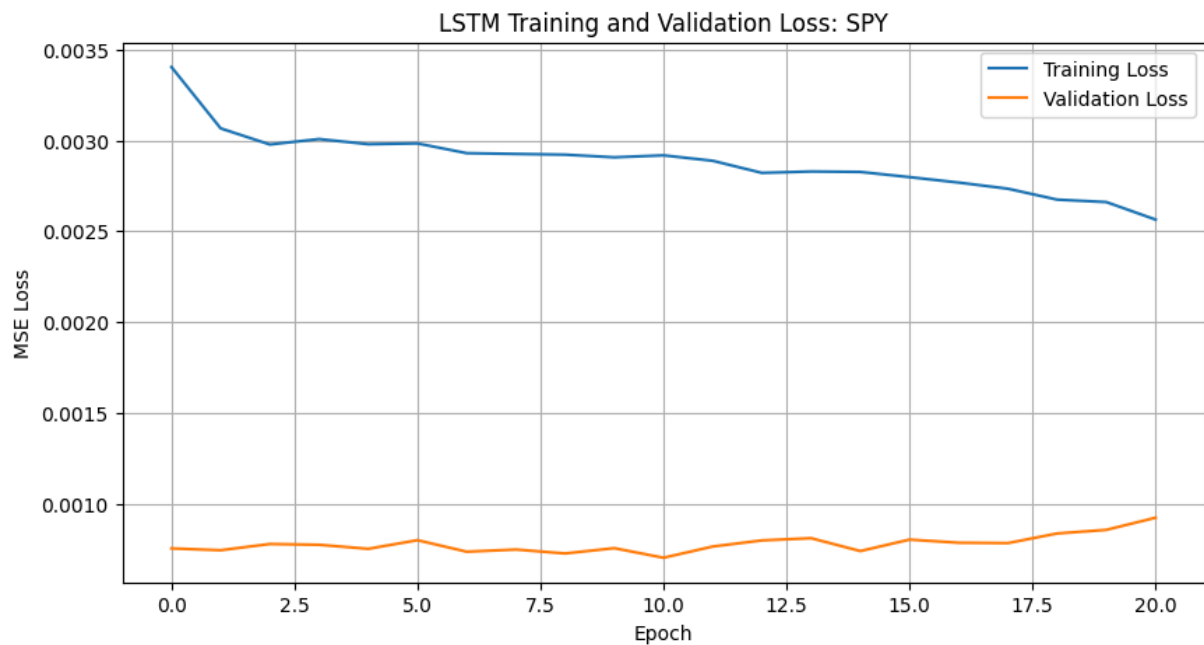


Figure 1.8. LSTM training and validation loss for SPY.

The SPY LSTM training loss changes slowly, while validation loss remains low. This suggests limited additional learning after early epochs.

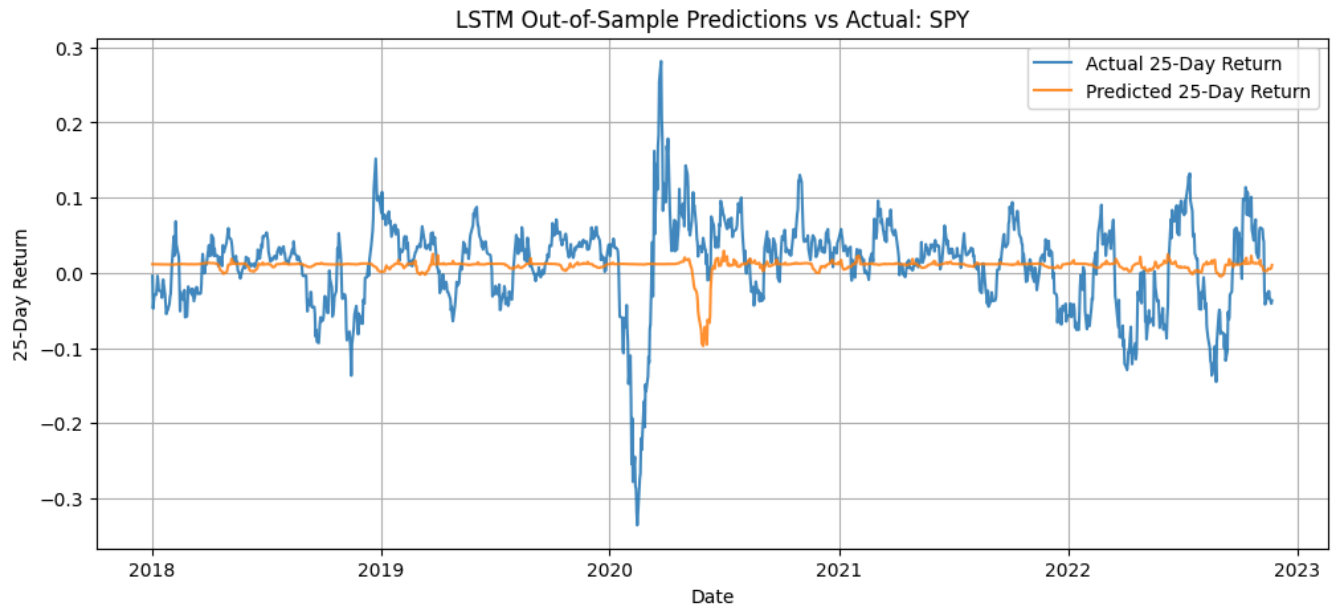


Figure 1.9. LSTM out-of-sample predictions for SPY.

The LSTM picks up some changes in direction, but it still understates the size of major equity moves. In practice, this means the model reacts to market changes, but not strongly enough to produce a sharp forecasting edge. This reflects the difficulty of forecasting equity returns from lagged returns alone.

Table 1.1. SPY out-of-sample performance across single-output models.

Model	Ticker	MSE	RMSE	MAE	Directional Accuracy	Prediction-Actual Correlation
MLP Test	SPY	0.003615	0.060126	0.042839	63.99%	0.161305
CNN-GAF Test	SPY	0.003559	0.059657	0.043665	66.99%	0.032748
LSTM Test	SPY	0.003669	0.060569	0.043806	65.13%	-0.025493

For SPY, CNN-GAF has the lowest RMSE, while CNN-GAF and LSTM have stronger directional accuracy than the MLP.

Even so, the differences across SPY models are not large, so the practical forecasting advantage remains modest. This suggests that model choice matters less than the overall difficulty of predicting short-horizon equity returns.

Section 2: TLT and SHY - Fixed-Income

TLT represents long-duration Treasury exposure, while SHY represents short-duration Treasury exposure. The pair helps distinguish interest-rate-sensitive bond risk from a more cash-like fixed-income asset.

2.1 Model Results for TLT

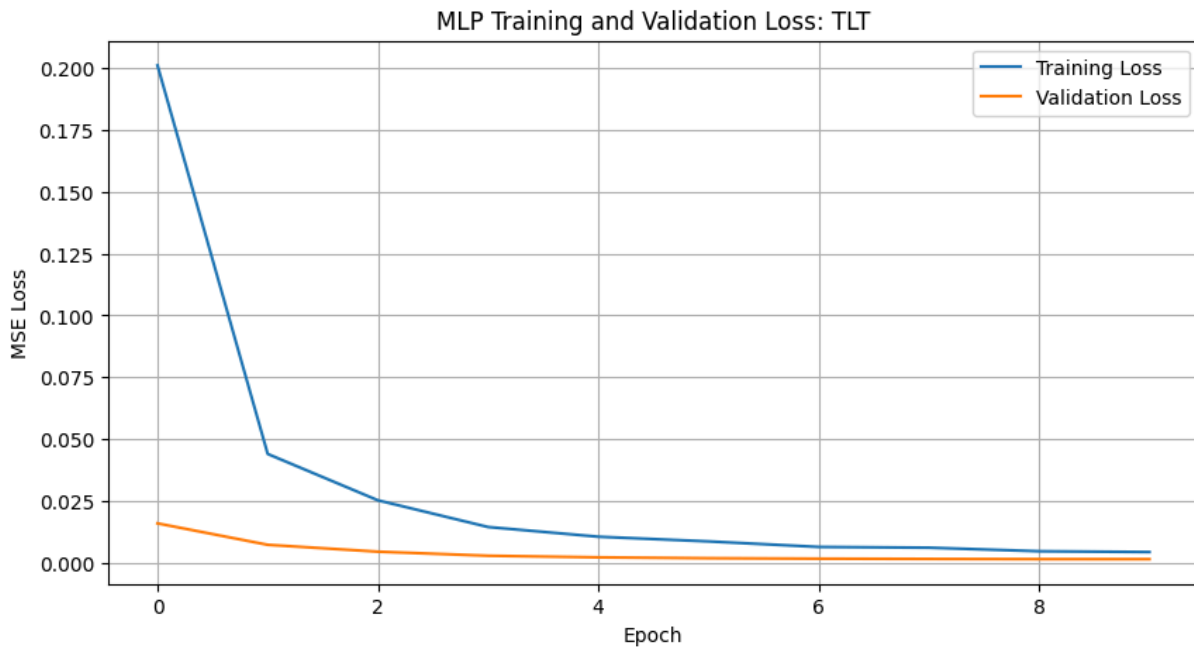


Figure 2.1. MLP training and validation loss for TLT.

The TLT MLP loss drops sharply early in training. Validation loss stays low, but prediction quality still depends on out-of-sample behaviour.

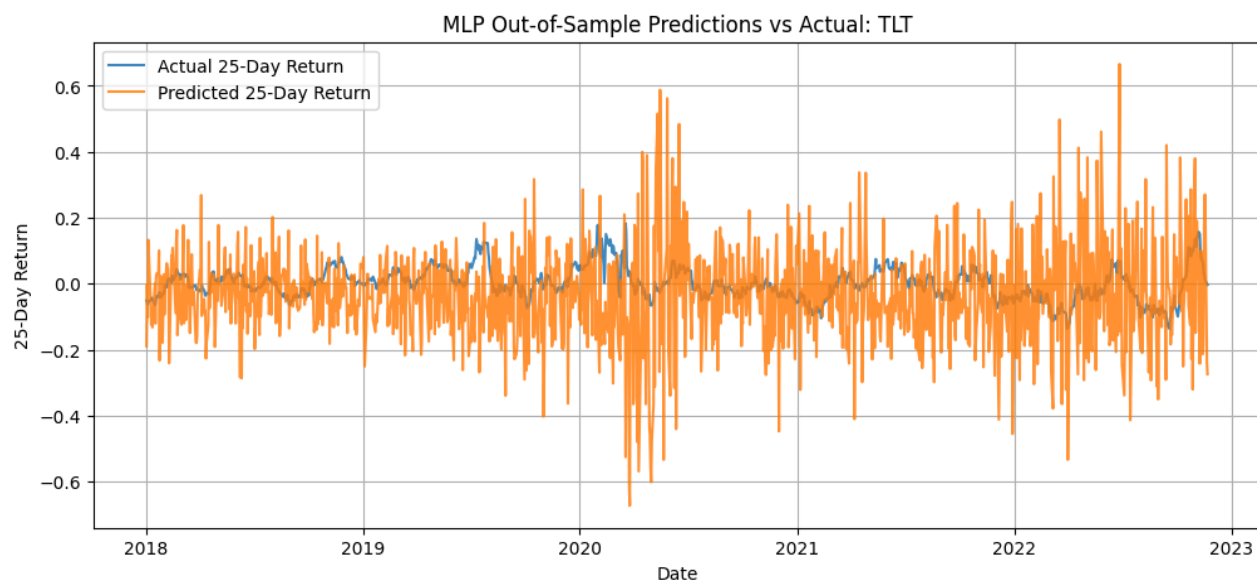


Figure 2.2. MLP out-of-sample predictions for TLT.

TLT MLP predictions are noisy and struggle with the size of bond-return swings. This weakens directional reliability.



Figure 2.3. Sample GAF image for TLT.

The TLT GAF image shows structured blocks created from the return sequence. The CNN uses this structure to search for bond-regime patterns.

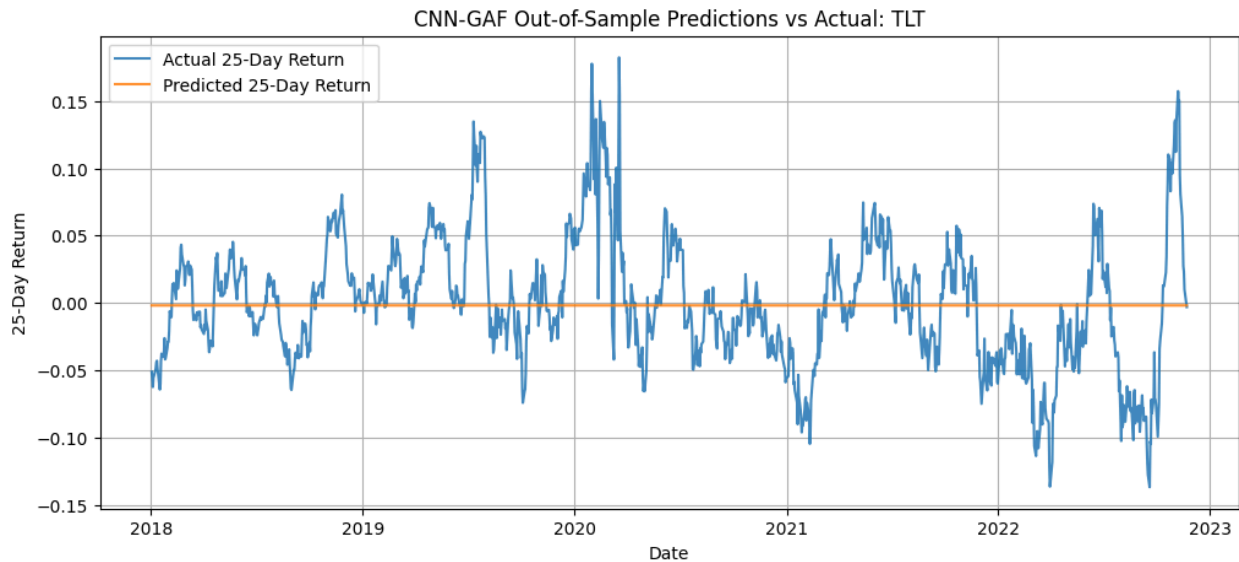


Figure 2.4. CNN-GAF out-of-sample predictions for TLT.

CNN-GAF predictions for TLT are smoother than the actual series. The model reduces forecast error compared with the TLT MLP.

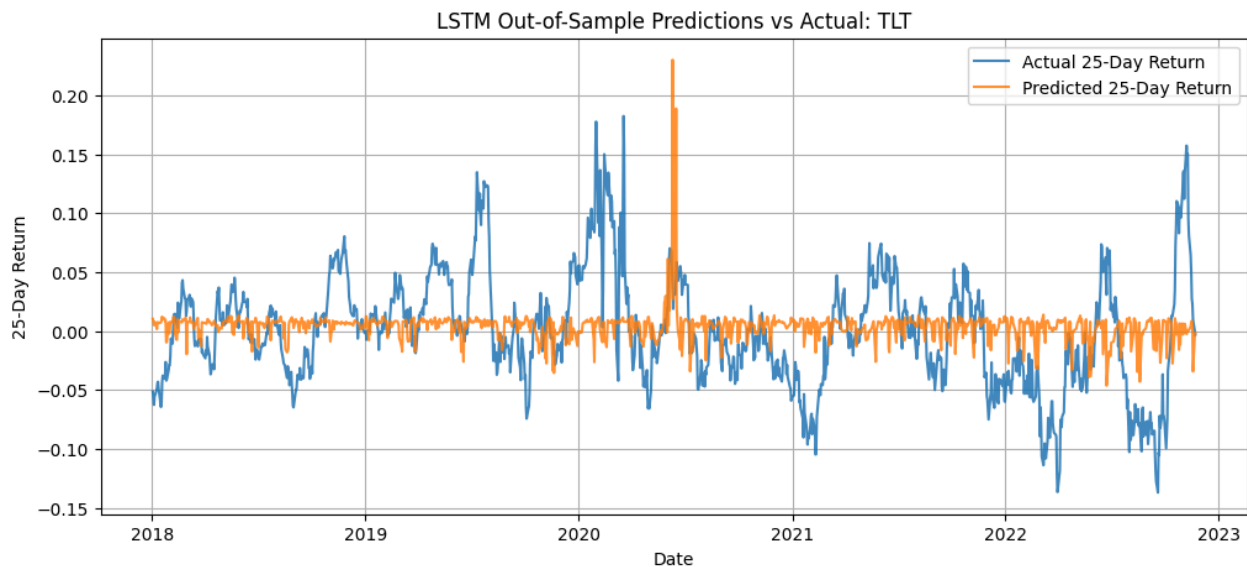


Figure 2.5. LSTM out-of-sample predictions for TLT.

The LSTM performs better than the MLP for TLT, but it still struggles when bond returns move abruptly. This is particularly important in rate-sensitive periods, when forecasting errors can quickly affect allocation decisions. Its correlation is positive, although modest.

2.2 Model Results for SHY

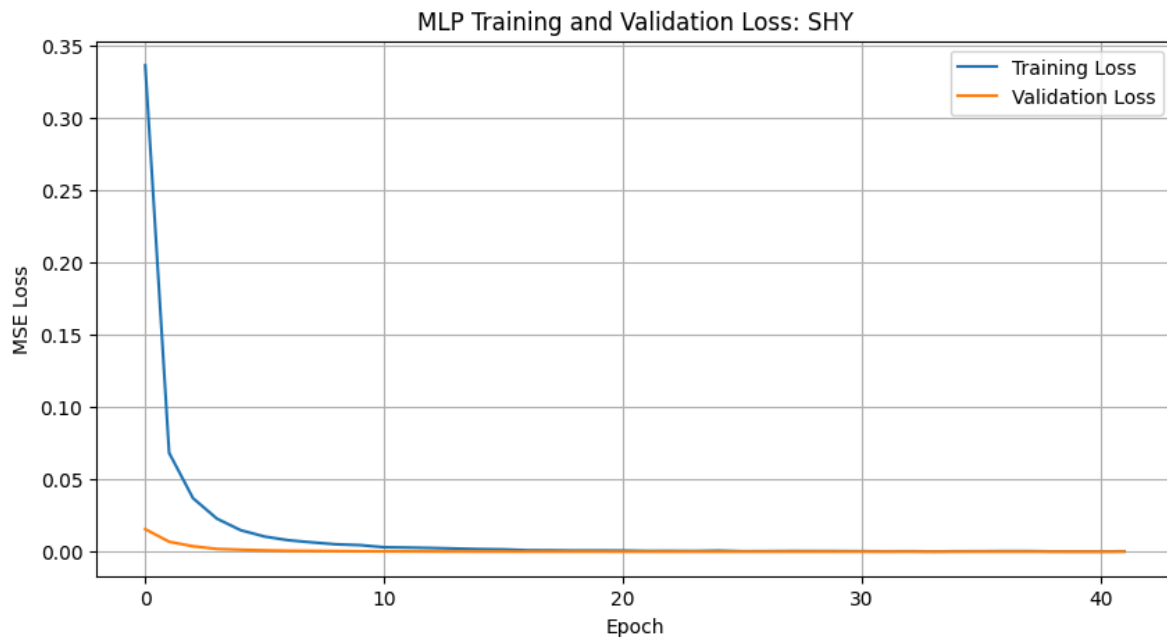


Figure 2.6. MLP training and validation loss for SHY.

SHY losses are very small because the asset has low volatility. The flat validation curve reflects the stability of short-duration Treasury returns.

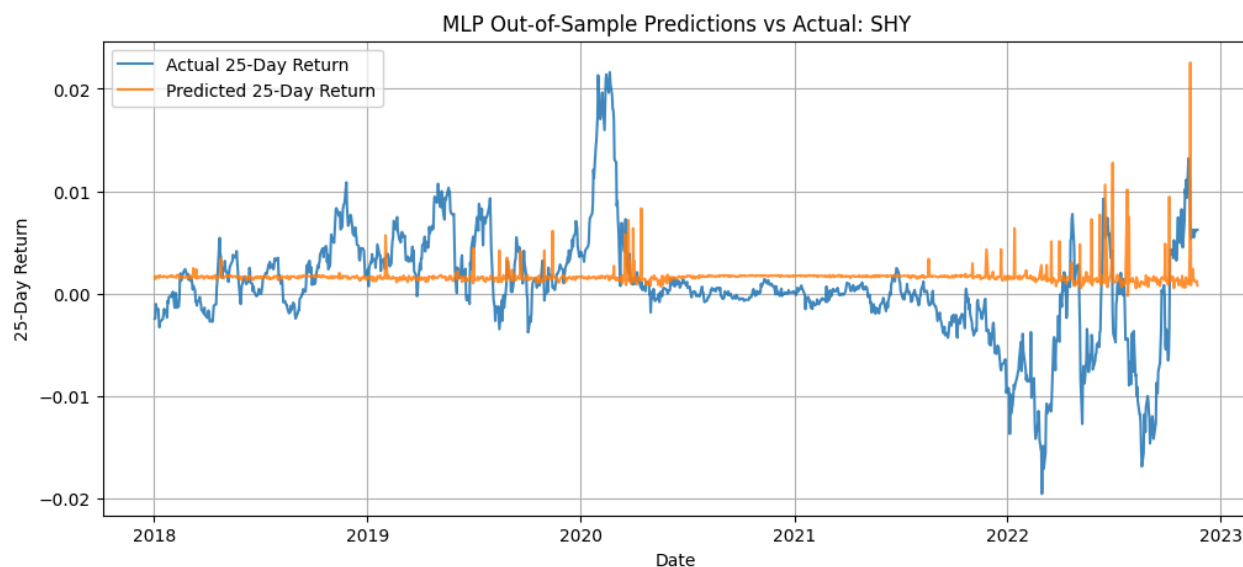


Figure 2.7. MLP out-of-sample predictions for SHY.

SHY predictions remain near zero, consistent with the asset's cash-like behaviour. Large directional signals are limited.

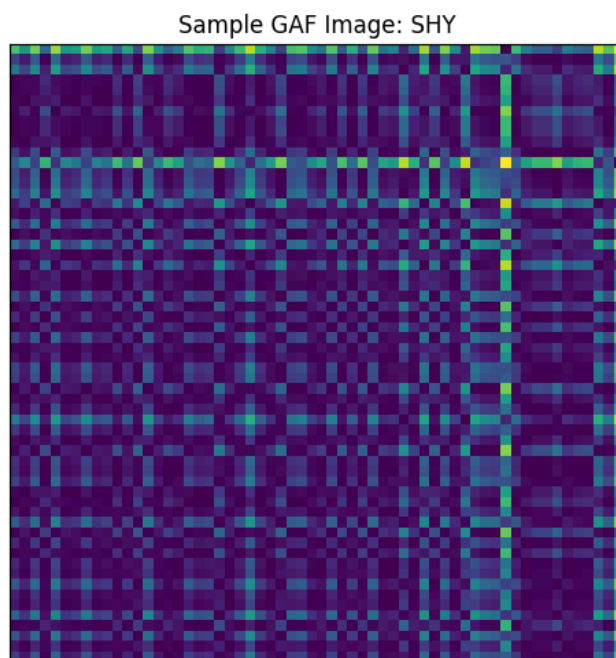


Figure 2.8. Sample GAF image for SHY.

The SHY GAF image still creates visual structure, but the underlying return series is much flatter. This limits the practical value of image-based modelling.

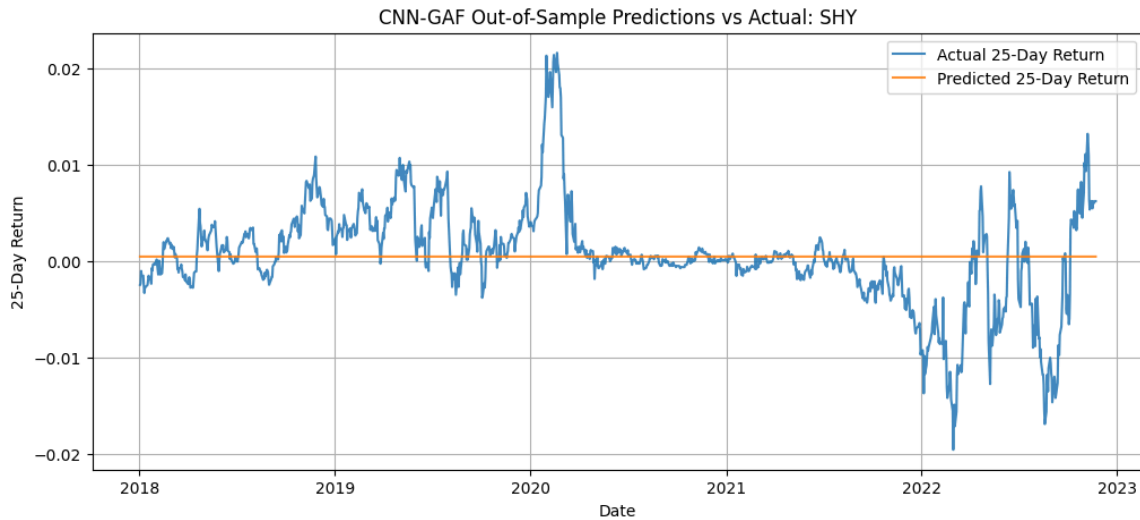


Figure 2.9. CNN-GAF out-of-sample predictions for SHY.

CNN-GAF predictions for SHY remain close to the low-volatility actual series. Forecast errors are small, but this mostly reflects SHY's stability.

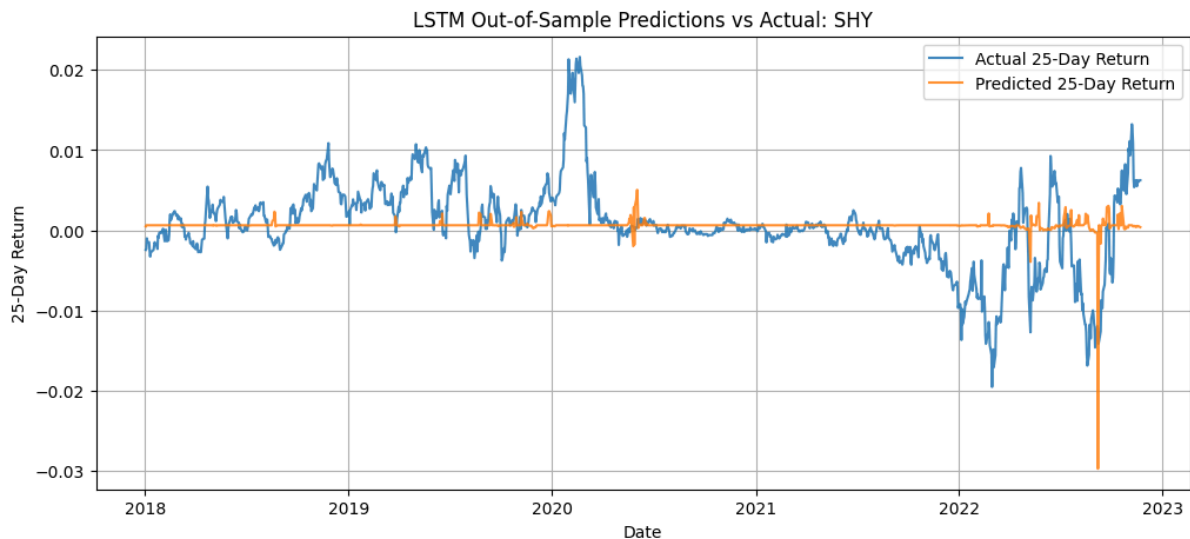


Figure 2.10. LSTM out-of-sample predictions for SHY.

For SHY, the LSTM mostly produces forecasts close to zero, which is consistent with the low-volatility nature of the asset. The low error is therefore partly a reflection of stability rather than strong predictive content. Its test correlation is positive, but the economic signal remains small.

Table 2.1. TLT and SHY out-of-sample performance across single-output models.

Model	Ticker	MSE	RMSE	MAE	Directional Accuracy	Prediction-Actual Correlation
MLP Test	TLT	0.025591	0.159972	0.122056	47.69%	-0.004743
MLP Test	SHY	2.90e-05	0.005369	0.003740	58.56%	-0.000616
CNN-GAF Test	TLT	0.002281	0.047757	0.036571	47.93%	N/A
CNN-GAF Test	SHY	0.000027	0.005181	0.003492	58.48%	-0.037297
LSTM Test	TLT	0.002343	0.048402	0.036760	51.58%	0.081193
LSTM Test	SHY	2.70e-05	0.005169	0.003473	59.21%	0.099732

The CNN-GAF correlation for TLT is not reported because the stored notebook output did not retain a valid prediction-actual correlation value for that test row. For SHY, all models show small errors because the asset is low-volatility.

2.3 Fixed-Income Strategy Interpretation

The TLT-SHY strategy highlights the importance of interest-rate regimes. Buy-and-hold TLT suffered large losses during 2022, while SHY remained comparatively stable because of its short-duration, cash-like behaviour. The rotation strategy did not outperform SHY because it still carried exposure to TLT during an unfavourable rate environment. This suggests that future allocation rules should use weighted positions, volatility controls, or regime filters instead of full-switch decisions.

Section 3: GLD and DBO - Commodity

GLD represents precious metals, while DBO represents crude oil exposure. These assets add commodity behaviour to the group model and help test whether deep learning can handle more volatile return dynamics.

3.1 GLD Results

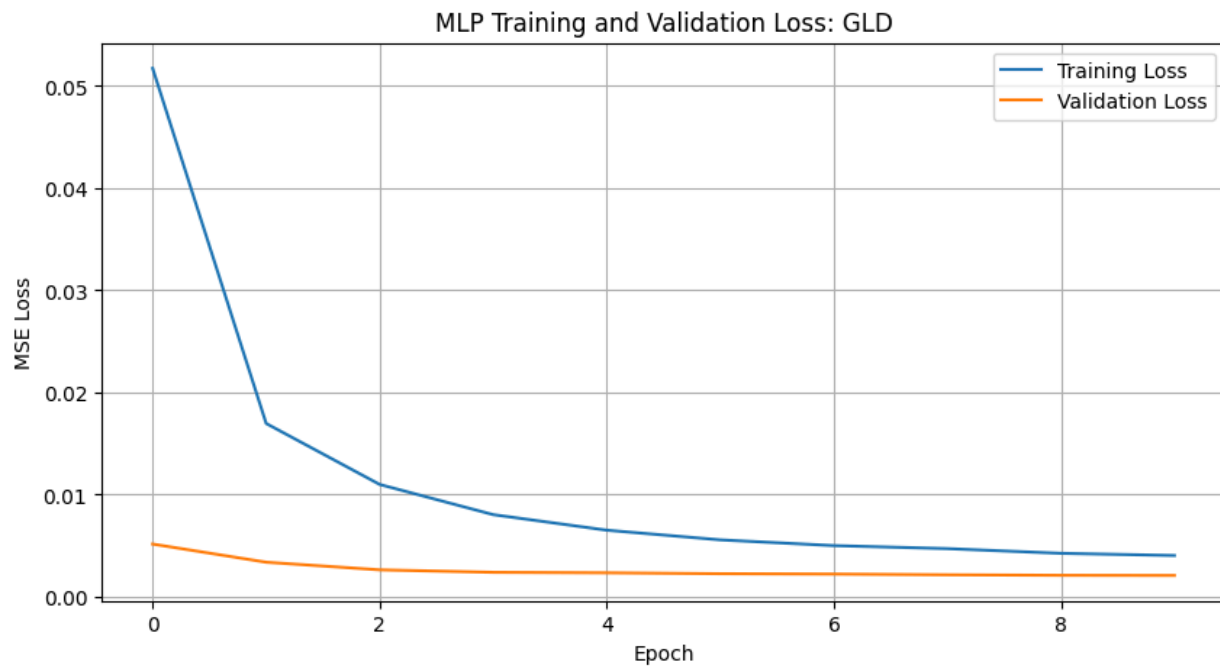


Figure 3.1. MLP training and validation loss for GLD.

The GLD MLP loss declines quickly and then flattens. This suggests early convergence with limited improvement after the first epochs.

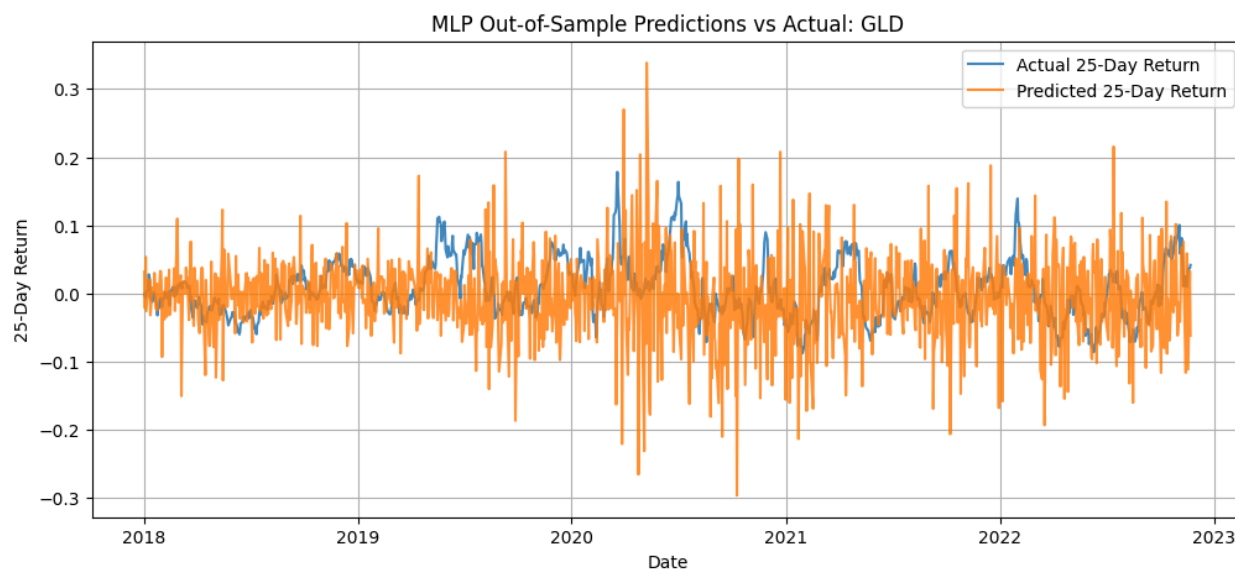


Figure 3.2. MLP out-of-sample predictions for GLD.

The GLD MLP predictions are smoother than actual returns. The model misses many large positive and negative 25-day moves.

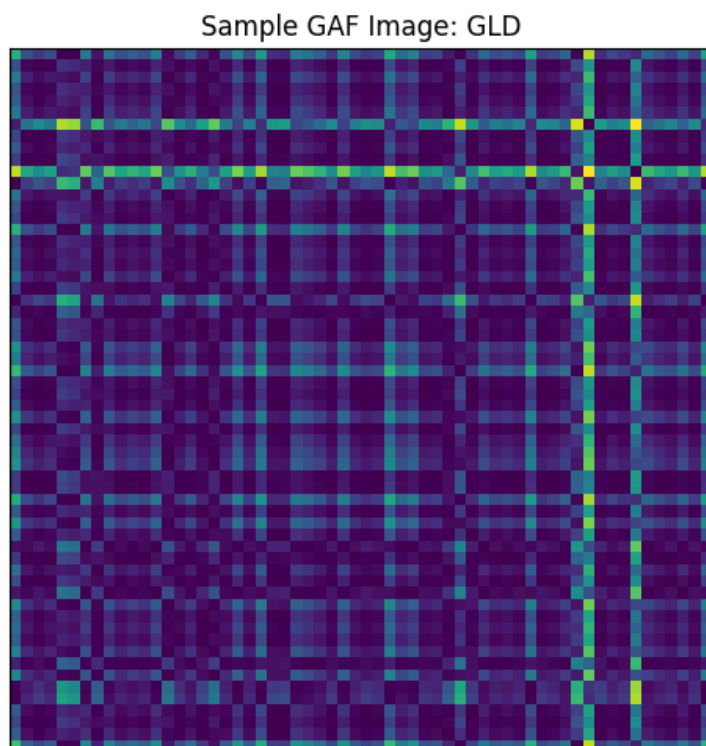


Figure 3.3. Sample GAF image for GLD.

The GLD GAF image translates a return sequence into a structured matrix. This allows the CNN to treat return history as an image-like input.

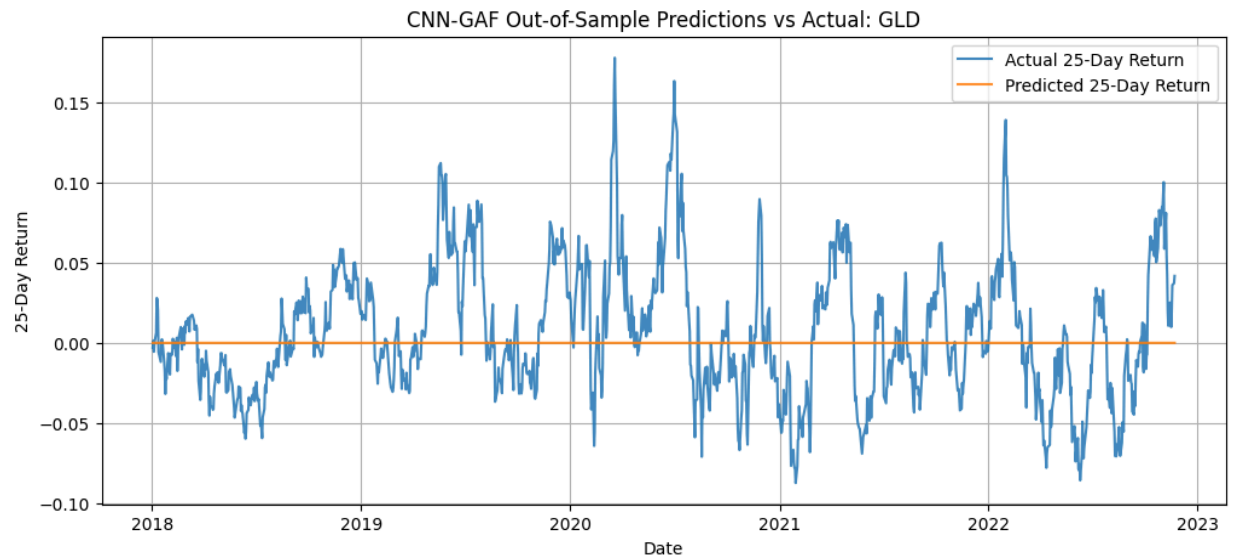


Figure 3.4. CNN-GAF out-of-sample predictions for GLD.

The CNN-GAF model remains conservative for GLD. Predictions stay near zero even when actual returns move sharply.

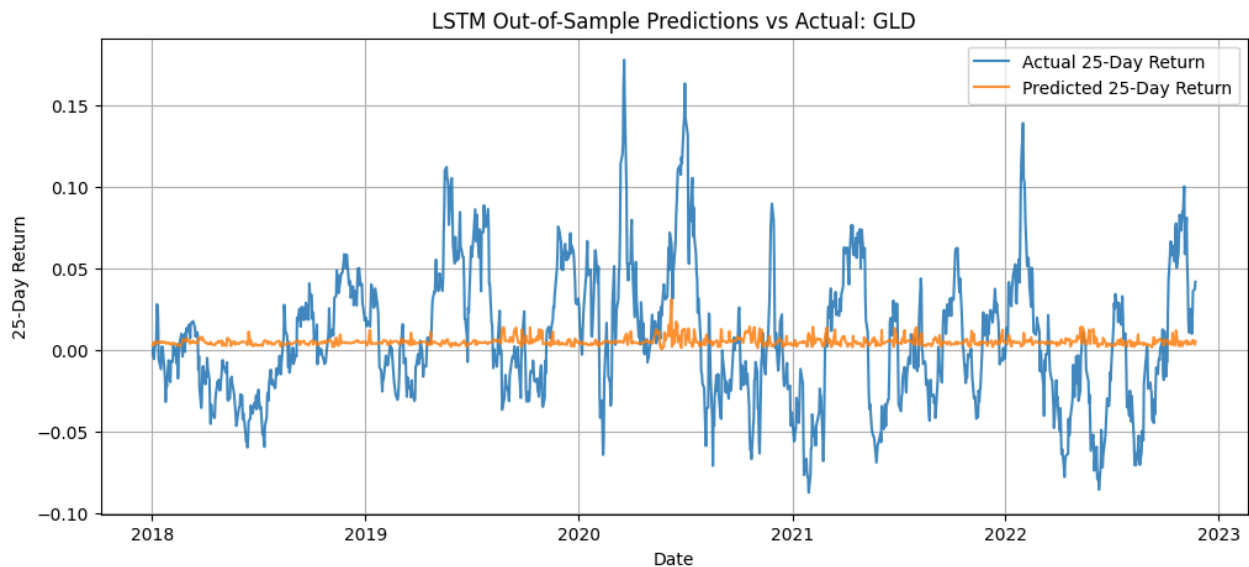


Figure 3.5. LSTM out-of-sample predictions for GLD.

For GLD, the LSTM does not materially improve on the earlier models. Its forecasts remain conservative, which suggests that short-horizon movements in gold were not strongly explained by lagged returns alone. The sequence model does not materially overcome the weak signal in GLD lagged returns.

3.2 DBO Results

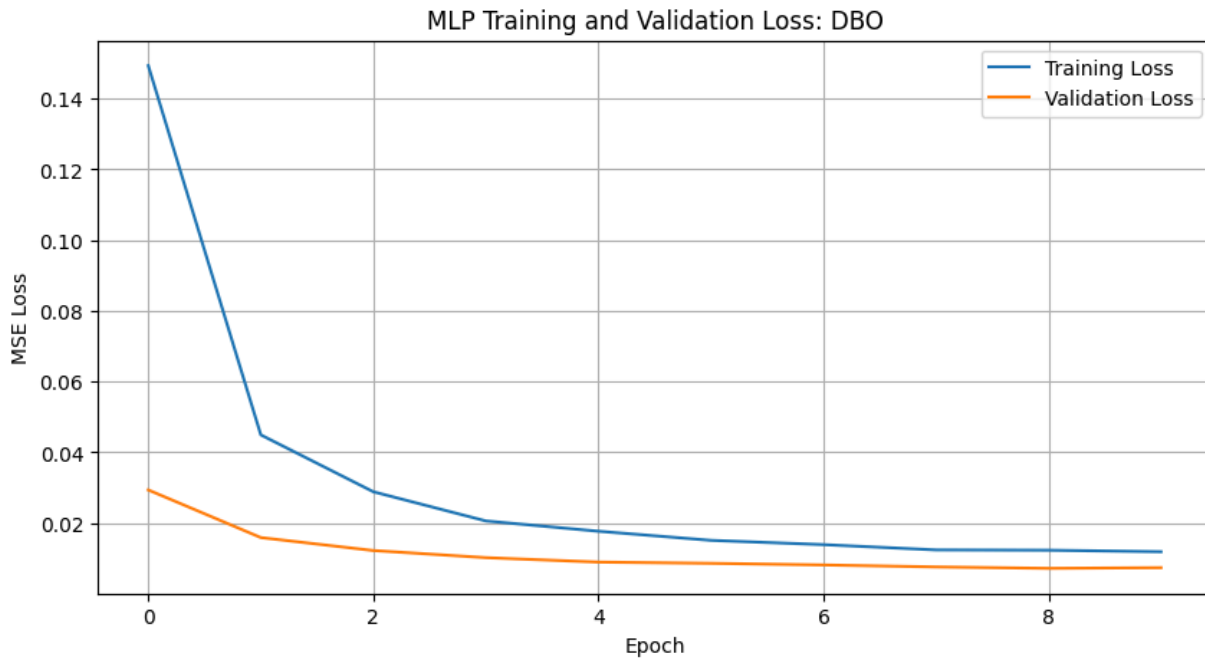


Figure 3.6. MLP training and validation loss for DBO.

DBO has higher loss levels than the lower-volatility assets. This reflects the difficulty of modelling crude-oil-linked returns.

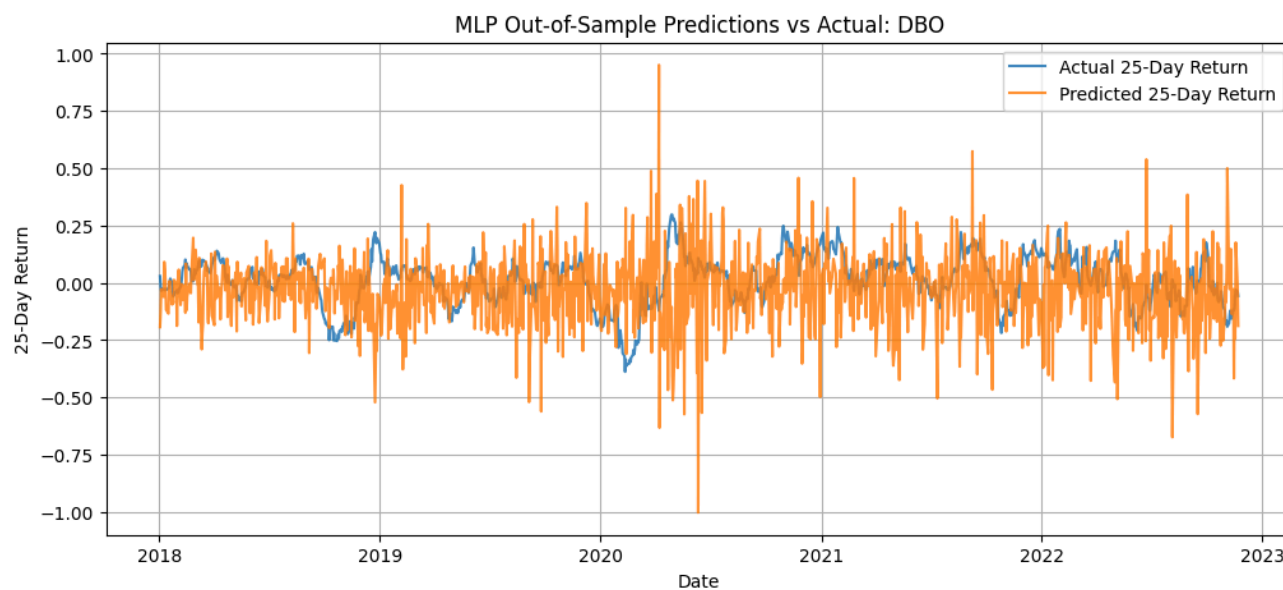


Figure 3.7. MLP out-of-sample predictions for DBO.

The DBO MLP struggles with large return swings. Its predictions are much smoother than the actual 25-day returns.

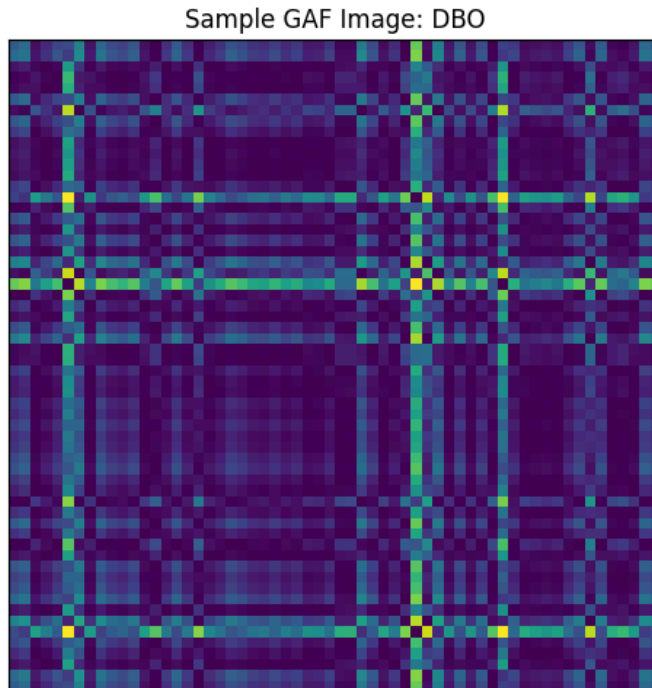


Figure 3.8. Sample GAF image for DBO.

The DBO GAF image shows strong contrast and block patterns. These patterns reflect larger movement in the crude-oil return window.

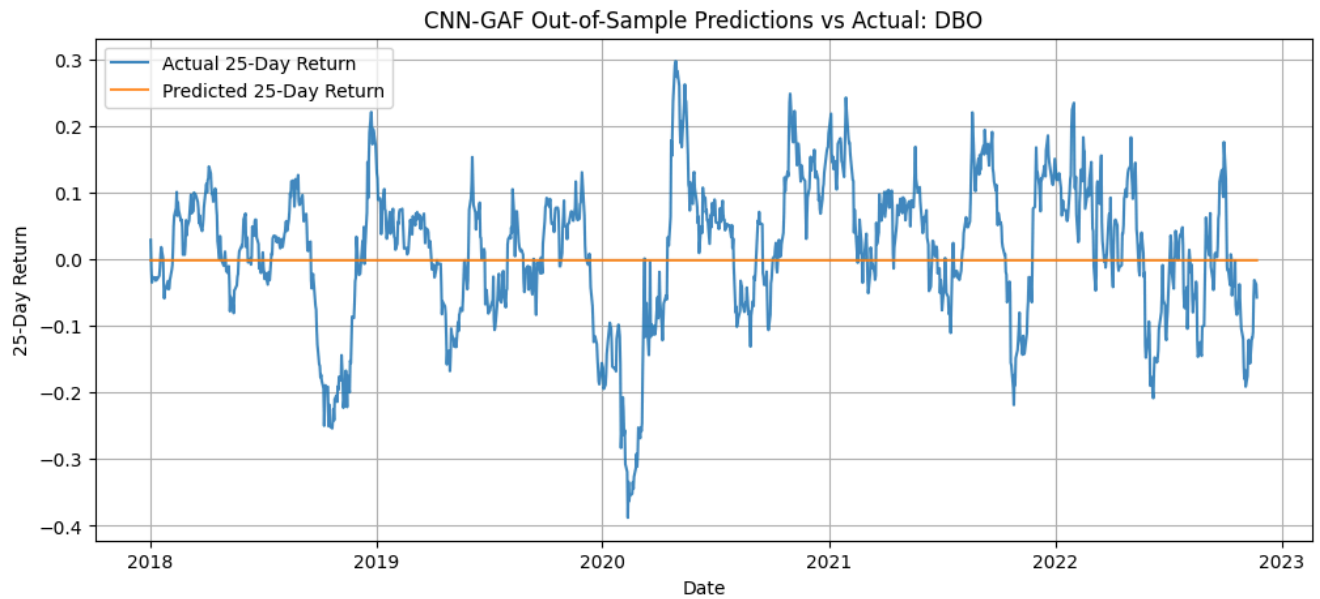


Figure 3.9. CNN-GAF out-of-sample predictions for DBO.

CNN-GAF improves DBO error relative to the MLP but still misses many large moves. Directional accuracy remains weak.

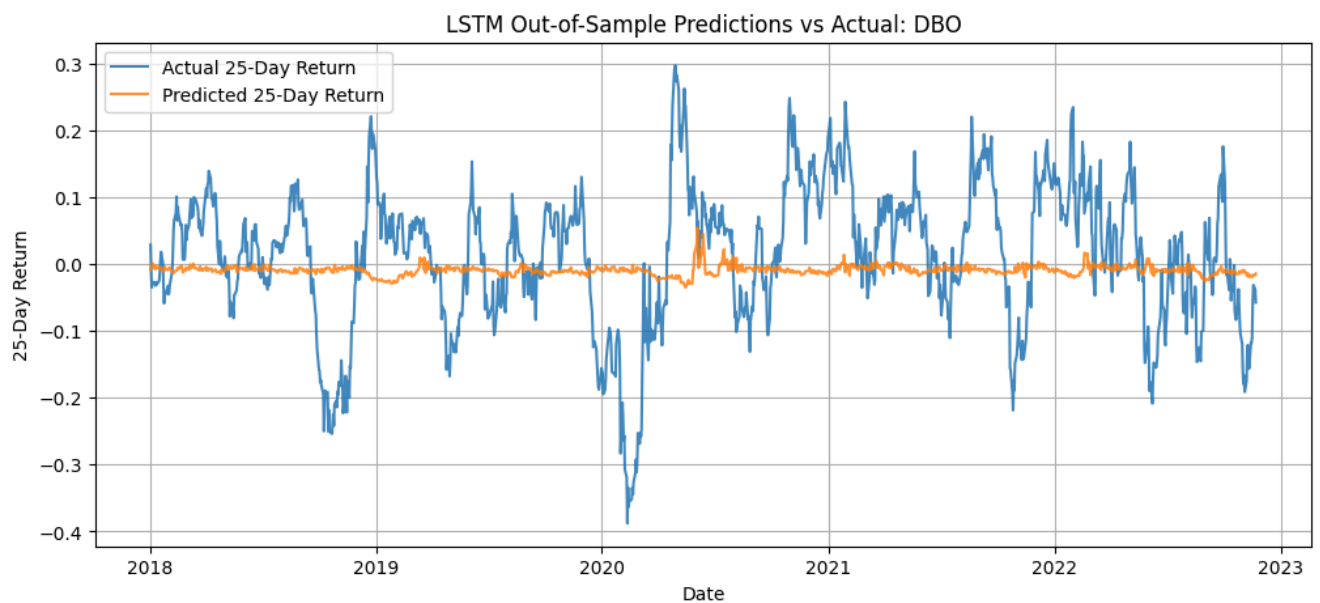


Figure 3.10. LSTM out-of-sample predictions for DBO.

For DBO, the LSTM reduces forecast error relative to the MLP, but the improvement is still limited from an economic perspective. The model handles normal periods better than extreme oil-price swings, which remain difficult to anticipate.

Table 3.1. GLD and DBO out-of-sample performance across single-output models.

Model	Ticker	MSE	RMSE	MAE	Directional Accuracy	Prediction-Actual Correlation
MLP Test	GLD	0.005646	0.075139	0.057270	49.63%	-0.007234
MLP Test	DBO	0.040153	0.200383	0.150427	49.39%	-0.036867
CNN-GAF Test	GLD	0.001700	0.041232	0.032154	51.42%	0.020313
CNN-GAF Test	DBO	0.011327	0.106428	0.083788	58.72%	-0.065720
LSTM Test	GLD	0.001686	0.041063	0.032403	51.42%	0.022822
LSTM Test	DBO	0.011835	0.108787	0.086354	44.20%	-0.077641

For GLD, all models have similar errors. For DBO, CNN-GAF and LSTM reduce RMSE substantially compared with the MLP.

The individual ETF sections contribute different asset-class perspectives to the final model. SPY provides the equity-market signal, TLT and SHY represent long- and short-duration fixed income, while GLD and DBO add commodity exposure. The consolidated model combines these signals into a shared multi-asset forecasting and allocation framework.

Taken together, the individual ETF results suggest that predictive strength varies across asset classes, with no single architecture dominating in every case. This motivates the final multi-output framework, which tests whether combining information across assets improves the overall signal.

Section 4: Consolidated Multi-Output Model and Allocation Strategy

The final team model uses all five ETF return histories together and predicts five 25-day ahead returns at once. This creates a common modelling layer after the individual ETF analyses.

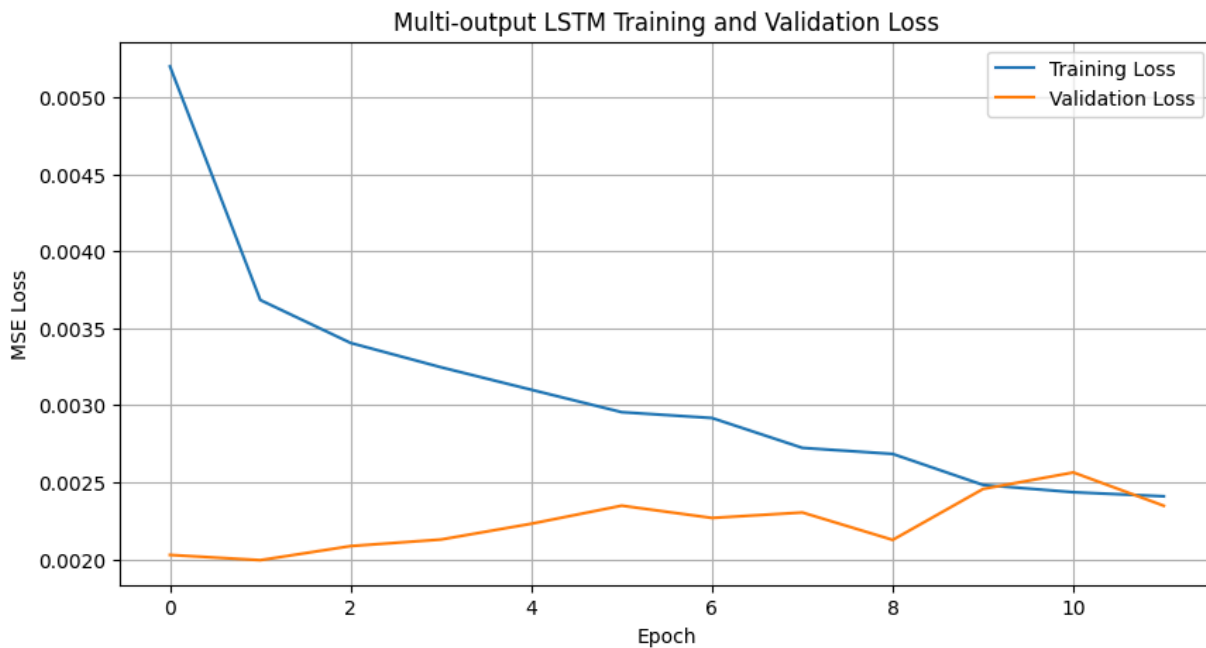


Figure 4.1. Multi-output LSTM training and validation loss.

Training loss declines steadily, while validation loss remains much lower and rises slightly later. This suggests improved fitting but limited validation improvement.

Table 4.1. Multi-output LSTM test performance across all ETFs.

Model	Ticker	MSE	RMSE	MAE	Directional Accuracy	Prediction-Actual Correlation
Multi-output LSTM Test	SPY	0.003796	0.061612	0.043144	65.53%	-0.095474
Multi-output LSTM Test	TLT	0.002415	0.049146	0.038084	48.42%	0.010107
Multi-output LSTM Test	SHY	6.70e-05	0.008204	0.004976	44.28%	0.042309
Multi-output LSTM Test	GLD	0.001665	0.040800	0.032529	49.31%	0.116018
Multi-output LSTM Test	DBO	0.011483	0.107159	0.084266	46.07%	0.054439

The multi-output LSTM has the lowest RMSE for SHY and moderate errors for SPY, TLT, GLD, and DBO. Correlations remain weak overall.

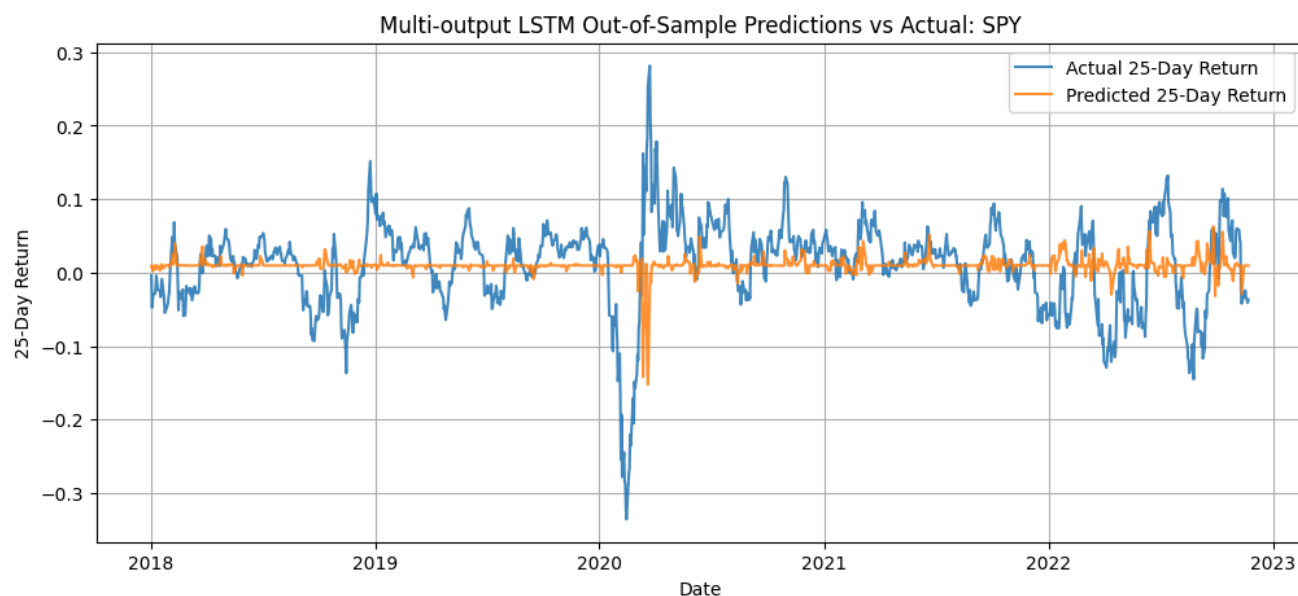


Figure 4.2. Multi-output LSTM predictions for SPY.

The multi-output SPY predictions remain smoother than actual returns. Directional accuracy is positive, but correlation is negative.

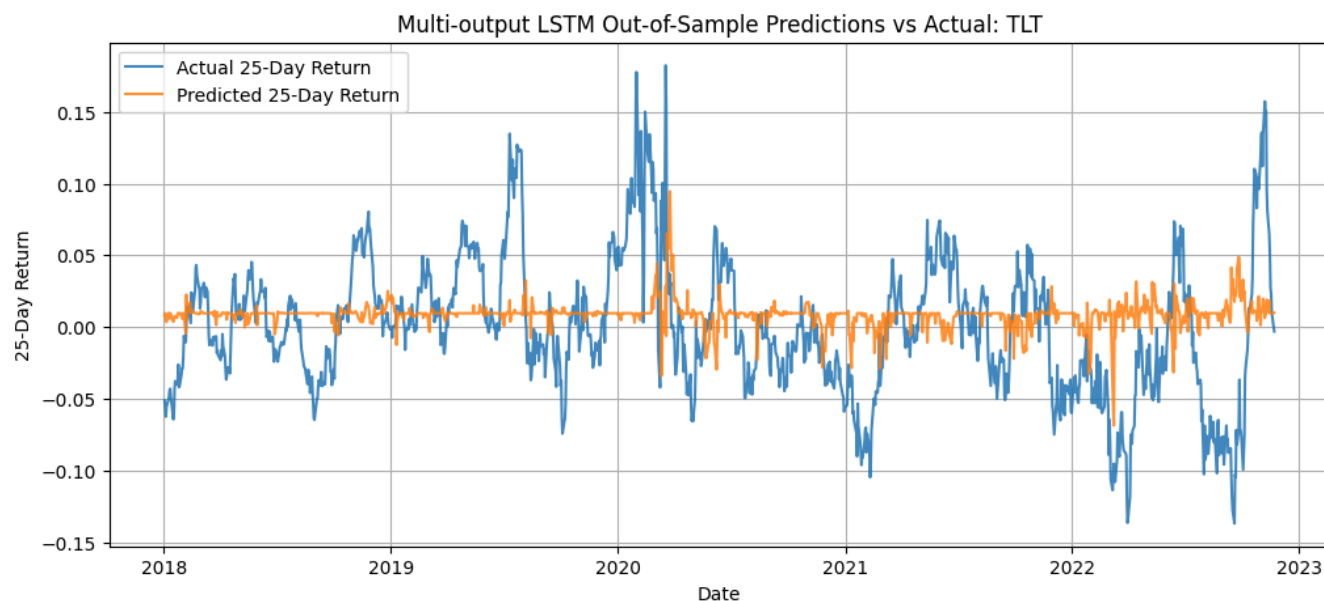


Figure 4.3. Multi-output LSTM predictions for TLT.

The multi-output TLT predictions are muted compared with actual returns. This limits their usefulness for timing bond-market swings.

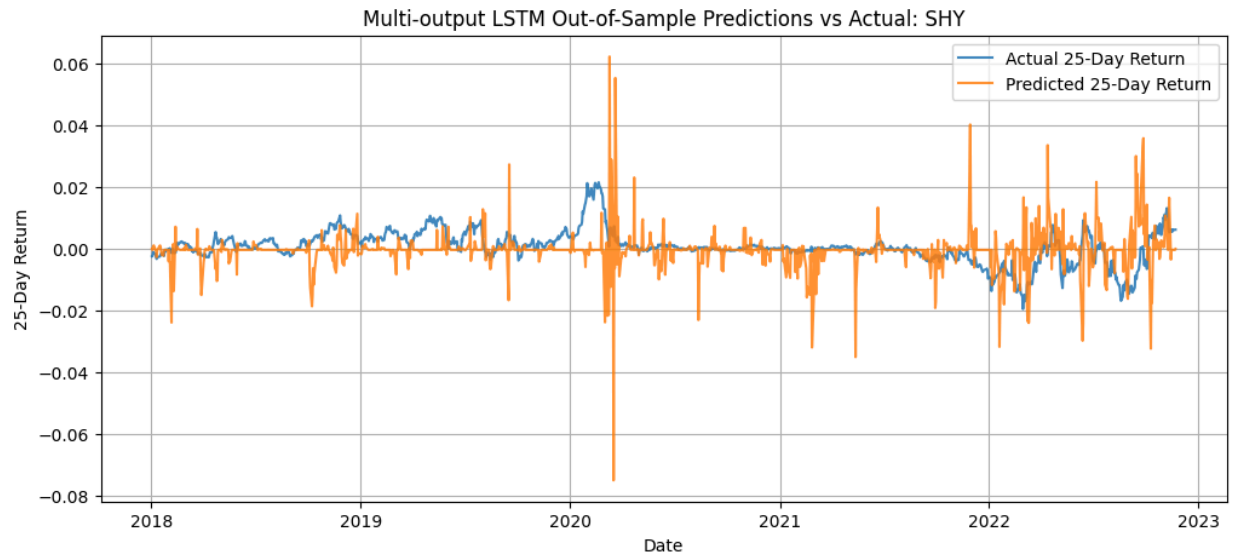


Figure 4.4. Multi-output LSTM predictions for SHY.

SHY predictions stay near zero because the actual series is stable. This produces small forecast errors but limited tactical signals.

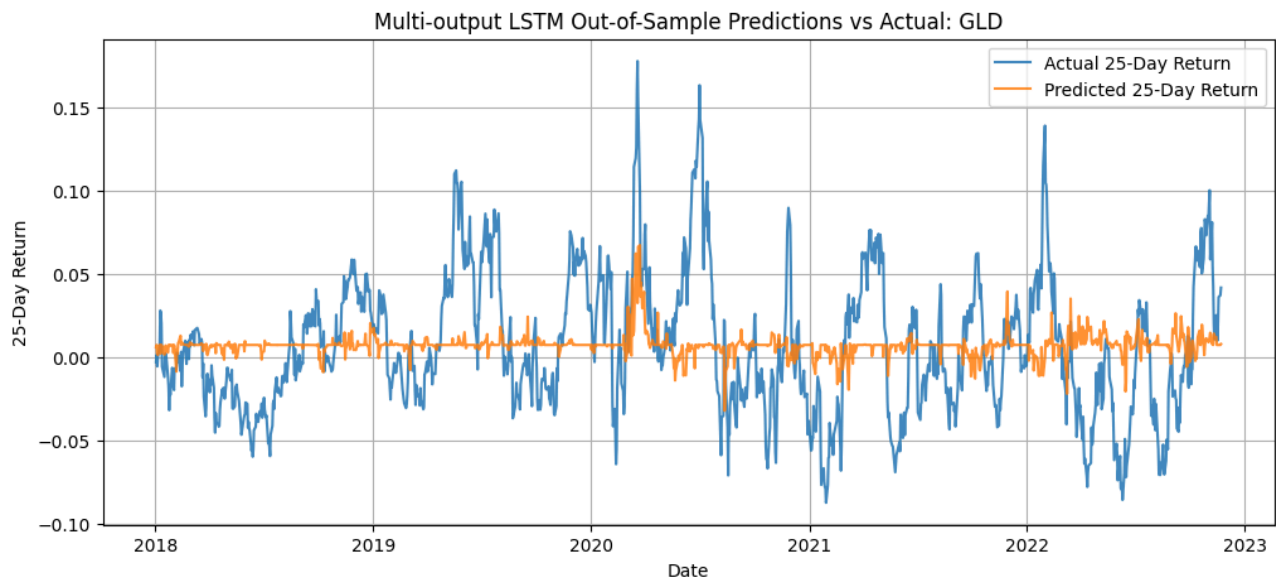


Figure 4.5. Multi-output LSTM predictions for GLD.

The multi-output GLD forecast is still smoother than actual returns. Correlation improves compared with the single-output LSTM.

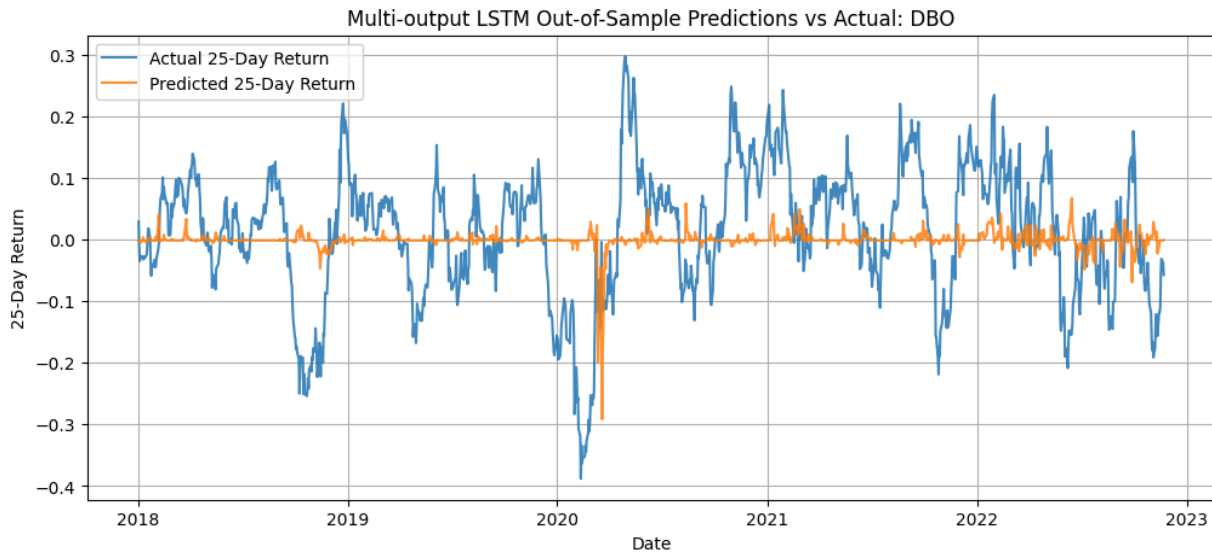


Figure 4.6. Multi-output LSTM predictions for DBO.

The DBO multi-output forecast captures limited direction but not the full size of major moves. Commodity volatility remains difficult to forecast.

Table 4.2. Single-output LSTM versus multi-output LSTM test comparison.

Model	Ticker	MSE	RMSE	MAE	Directional Accuracy	Prediction-Actual Correlation
LSTM Test	SPY	0.003669	0.060569	0.043806	65.13%	-0.025493
LSTM Test	TLT	0.002343	0.048402	0.036760	51.58%	0.081193
LSTM Test	SHY	2.70e-05	0.005169	0.003473	59.21%	0.099732
LSTM Test	GLD	0.001686	0.041063	0.032403	51.42%	0.022822
LSTM Test	DBO	0.011835	0.108787	0.086354	44.20%	-0.077641
Multi-output LSTM Test	SPY	0.003796	0.061612	0.043144	65.53%	-0.095474
Multi-output LSTM Test	TLT	0.002415	0.049146	0.038084	48.42%	0.010107
Multi-output LSTM Test	SHY	6.70e-05	0.008204	0.004976	44.28%	0.042309
Multi-output LSTM Test	GLD	0.001665	0.040800	0.032529	49.31%	0.116018
Multi-output LSTM Test	DBO	0.011483	0.107159	0.084266	46.07%	0.054439

The multi-output model improves correlation for GLD and DBO, but not for all ETFs. Cross-asset information helps unevenly across the portfolio.

4.1 Asset Allocation Backtest

The allocation rule rebalances every 25 trading days. At each rebalance, the strategy selects the two ETFs with the highest predicted 25-day returns and assigns 50% to each. The benchmark is an equal-weight buy-and-hold portfolio.

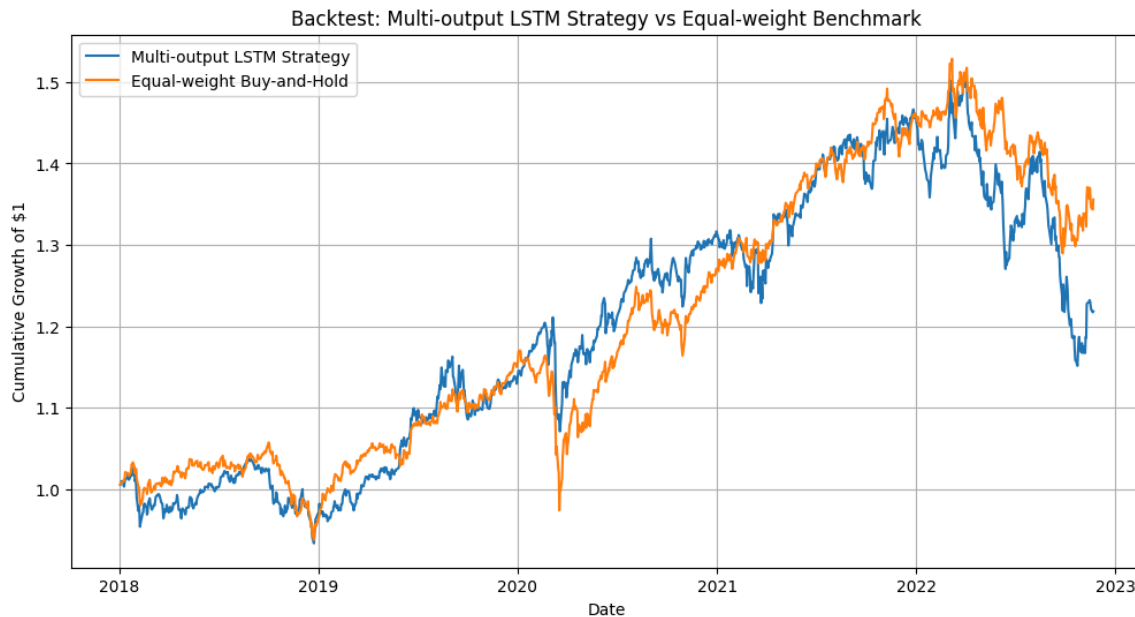


Figure 4.7. Multi-output LSTM strategy versus equal-weight benchmark.

The equal-weight benchmark outperforms the model-driven strategy over the test period. The strategy participates in growth but experiences weaker cumulative performance.

Table 4.3. Backtest performance metrics.

Strategy	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Multi-output LSTM Strategy	21.84%	4.12%	11.75%	0.351	-23.44%
Equal-weight Buy-and-Hold	35.57%	6.42%	10.09%	0.636	-16.78%

The benchmark has higher cumulative return, higher annualized return, higher Sharpe ratio, and lower maximum drawdown than the model-driven strategy.

In other words, the model was often able to rank assets, but not with enough conviction or timing accuracy to outperform a simple diversified benchmark. The early allocations also favoured SPY and TLT repeatedly, which reduced diversification and exposed the strategy to bond-market weakness.

Table 4.4. First ten rebalance allocations.

Date	SPY_weight	TLT_weight	SHY_weight	GLD_weight	DBO_weight
2018-01-02	0.500000	0.500000	0.000000	0.000000	0.000000
2018-02-07	0.500000	0.500000	0.000000	0.000000	0.000000
2018-03-15	0.500000	0.500000	0.000000	0.000000	0.000000
2018-04-20	0.500000	0.500000	0.000000	0.000000	0.000000
2018-05-25	0.500000	0.500000	0.000000	0.000000	0.000000

2018-07-02	0.500000	0.500000	0.000000	0.000000	0.000000
2018-08-07	0.500000	0.500000	0.000000	0.000000	0.000000
2018-09-12	0.500000	0.500000	0.000000	0.000000	0.000000
2018-10-17	0.500000	0.500000	0.000000	0.000000	0.000000
2018-11-21	0.500000	0.000000	0.000000	0.500000	0.000000

The early allocation favours SPY and TLT, then later introduces GLD. This shows how the strategy converts model forecasts into portfolio weights.

The repeated early allocation to SPY and TLT shows that the model concentrated in a narrow part of the asset universe. This helped the strategy participate in equity upside but also limited the defensive benefit of SHY and the diversification potential of GLD and DBO.

Section 5: Discussion and Conclusion

Across the models, in-sample fitting was generally stable, but out-of-sample forecasts were often too smooth and missed sharp return movements. Performance also varied by ETF: SHY had low errors because of its stability, while DBO remained harder to predict because of higher volatility.

The multi-output LSTM captured some cross-asset information, especially for GLD and DBO, but the improvement was not consistent across all assets. CNN-GAF improved directional accuracy for SPY and DBO, but correlations remained weak, showing that the models still struggled to track the size of actual returns.

Overall, deep learning helped structure return information, but lagged returns alone were not strong enough to support consistent portfolio outperformance. Future improvements should include macroeconomic variables, volatility features, transaction costs, and risk-adjusted weighting.

References

- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427-431.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357-384.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780.
- LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.
- Wang, Z., & Oates, T. (2015). Imaging time-series to improve classification and imputation. *Proceedings of the 24th International Joint Conference on Artificial Intelligence*, 3939-3945.
- Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). *Dive into Deep Learning*. Cambridge University Press.